

MA 301
Test 2
Spring - 2015

Problem	points	out of
1		6
2		5
3		10
4		9
Score		30

Name: _____

HR: _____

Signature: _____

READ BEFORE YOU START
* Show all work. * Change your calculator mode to RADIANS. * Sloppy solutions may be penalized. * Use calculators ONLY for computations. * You must write up every problem in a clear and organized way. * If you write solutions on a page different than the question, then clearly state where it is located.

Problem 1: Suppose $\mathbf{F} = \left[\cos(\pi x), xz, \frac{1}{2}x^2 + \ln y \right]$ and S the surface: $z = 4 - y^2$ where $y \geq 0$ and $0 \leq x \leq 3$ with outward pointing normal (away from the x -axis). Let C be the curve which bounds S .

(a) Compute the line integral of $\mathbf{F}$ on C : $\oint_C \mathbf{F} \cdot d\mathbf{r}$

(b) Set-up the double integral that computes the flux of $\mathbf{F}$ through S : $\iint_S \mathbf{F} \cdot \mathbf{n} \, dS$

Problem 2: Let $\mathbf{F} = [107, ze^{xz}, xyz]$ and S be the surface (with outward orientation) of the solid bounded by the surfaces $x = 0$, $x = 5$, $y = 0$, $z = 0$, and $y^2 + z^2 = 1$ in the first octant.

Compute $\iint_S \mathbf{F} \cdot \mathbf{n} dS$

Problem 3: Let $z_1 = -3i$, $z_2 = 3 - 3i$, $z_3 = -5$, $z_4 = -\sqrt{3} - i$. (Do NOT leave any answers as decimals.)

****Bonus:** 2 additional bonus points will be given if you do all work using Euler's Polar Form (i.e. $z = re^{\theta i}$) on (b) - (d).

(a) Perform the following operations and leave answer in standard form.

▷ $z_1 - z_4$

▷ $\frac{z_4}{z_2}$

(b) Write z_1, z_2, z_3, z_4 in polar form.

▷ $z_1 =$

▷ $z_2 =$

▷ $z_3 =$

▷ $z_4 =$

(c) Perform the following operations and leave answer in polar form.

▷ $z_1 z_2$

▷ $(z_1 z_2)^6$

(d) Solve the following complex equations for x and leave your solutions in polar form:

▷ $\frac{x^{1/6}}{z_1} - z_2 = 0.$

▷ $x^3 - z_3 = 0.$

Problem 4: Answer the following quick computation questions:

(a) Let $\mathbf{F} = -y\mathbf{i} + x\mathbf{j} + e^z\mathbf{k}$ and S be a circle of radius 3 inside the plane $x + 2y + z = 10$. If C is the boundary of this circle (oriented counterclockwise) then find $\oint_C \mathbf{F} \cdot d\mathbf{r}$ (your answer should be in terms of $a, b, \& c$)

(b) Compute $\oint_C \mathbf{F} \cdot d\mathbf{r}$ where $\mathbf{F} = [5y, 2z, 5x]$ and C is the boundary around the triangle obtained by tracing out the path $(2, 0, 0)$ to $(2, 0, 2)$, to $(2, 2, 2)$ back to $(2, 0, 0)$.

(c) Compute $\iint_S \mathbf{F} \cdot \mathbf{n} dS$ where $\mathbf{F} = [5x^2 + y, -3xy - 3y, -6xz]$ and S is the surface of the solid cube with $0 \leq x \leq 2$, $|y| \leq 1$, and $|z| \leq 1$ with outward pointing normal.